

Minimum Wage Many-Analyst Experiment

Executive Summary and Project Details

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1 Executive summary

This project tests whether **prompted literature priors** causally change what autonomous AI agents report about the employment effects of the minimum wage. Agents analyze the **same** state-year panel with the **same** task instructions; only the literature context (if any) is randomized. Multiple AI models are tested to examine whether model choice mediates the influence of literature primes.

Design. Five arms $\times$ 50 Cursor agents per model. Arms: no prime; null-effect short; null-effect with citations; negative-effect short; negative-effect with citations. Agents choose outcome, sample window, treatment definition, covariates, and estimator under a shared DiD methodology guide.

Headline results — Composer-2 run (Second Step, 250 agents).

Finding	Result
Completion	250/250 agents returned usable results.csv
Between-arm ATT differences	Not significant (Fisher omnibus $p = 0.4101$; Kruskal-Wallis $p = 0.3063$)
Outcome choice	~99.6% studied teen employment-to-population
Estimator choice	~98.8% Callaway-Sant’Anna family; 4% TWFE in any arm
Treatment definition	249/250 binary; 0% continuous MW treatment
Short vs citation-rich primes	No meaningful ATT gap on planned contrasts

Headline results — Opus 4.6 run (Third Step, 249 agents, 1 failure).

Finding	Result
Completion	249/250 agents returned usable results.csv
Between-arm ATT differences	Significant (Fisher omnibus $p = 0.0010$; Kruskal-Wallis $p = 0.0073$)
Outlier arm	negative_short : mean ATT = -0.0027, median = -0.0085 (most negative)
Specification variation	negative_short shows 3 continuous treatment, 2 CGBS, 1 TWFE — more diverse than other arms
Pairwise	null_cites vs negative_short $p = \text{NA}$

Pending run. A **Grok 4.6** run (250 agents) is currently in progress; results will be incorporated when complete.

Bottom line. With Cursor **composer-2** agents, literature primes did **not** shift reported ATTs — behavior resembled Cunningham’s constrained **Wave 1**. With **Opus 4.6**, the **negative_short** arm produced significantly more negative ATTs and more specification diversity, **more closely resembling Cunningham’s Wave 2** finding that negatively primed agents moved toward alternative estimators and more negative estimates. **Model choice is a critical moderator** of whether literature primes influence agent behavior.

Implication. The same experimental protocol produces qualitatively different results across models: primed priors alone are not enough with **composer-2**, but **Opus 4.6 agents respond to negative primes** through both ATT shifts and specification-channel divergence. Platform and model matter.

2 1. Purpose and motivation

Many-analyst studies show that researchers facing identical data can report different causal estimates through **specification choices**. Borjas & Breznau (2026, *Science Advances*) linked immigration researchers’ priors to their estimates. Scott Cunningham’s design turns that association into an **experiment** by randomizing the prior via a literature prime while holding data and task fixed ([Substack write-up](#)).

This repository is a **Cursor SDK replication / extension** of Cunningham’s `mw-analyst-experiment` design (originally Claude Code CLI).

Research question: Do literature primes about minimum-wage employment effects causally shift autonomous agents’ reported average treatment effects (ATTs)?

3 2. Experimental design

3.1 Arms

Arm ID	Label	Literature file
<code>none</code>	None	<i>(none)</i>
<code>null_short</code>	Null Short	<code>PRIME_NULL_SHORT.md</code>
<code>null_cites</code>	Null Cites	<code>PRIME_NULL_CITES.md</code>
<code>negative_short</code>	Negative Short	<code>PRIME_NEGATIVE_SHORT.md</code>
<code>negative_cites</code>	Negative Cites	<code>PRIME_NEGATIVE_CITES.md</code>

All agents receive: `agent_panel_essential.csv`, `DATA_DICTIONARY.md`, `DID_METHODODOLOGY.md`, and `INSTRUCTIONS_SHARED.md`. Primed arms also receive one literature context file.

3.2 What agents decide

- Employment outcome (CPS subgroups, QCEW industries, etc.)
- Treatment definition (binary staggered vs continuous)
- Sample years / states
- Covariates and comparison group
- Estimator (Callaway–Sant’Anna, continuous DiD, TWFE, etc.)

3.3 Deliverables per agent

1. `results.csv` — one row with design choices and ATT / SE / CI
2. `llms.txt` — structured narrative of the analysis

3.4 Data

Pre-aggregated **state** × **year** panel (51 units, 1990–2022): effective minimum wages, CPS employment rates for multiple groups, and selected QCEW industry series.

3.5 Scale and waves

Wave	Location	Scale	Model	Notes
First	First Step/	150 agents (30×5)	composer-2	Earlier protocol; TWFE discouraged
Second	Second Step/results/	250 agents (50×5)	composer-2	TWFE allowed in methodology
Third	Third Step/results_opus46/	249 agents (50×5, 1 failure)	opus-4.6	Same protocol as Second; different model
Fourth (in progress)	Third Step/results_grok46/	250 agents (50×5)	grok-4.6	Same protocol; pending completion

4 3. Technical pipeline

Component	Role
@cursor/sdk + launch_experiment_cursor.mjs	Local agents in isolated temp workspaces
run_full_experiment.sh / run_pilot_one_per_condition.sh	Full (250) and pilot (5) launchers
collect_results.sh	Aggregate outputs → Second Step/results/
analyze_results.R	Omnibus tests, plots
analyze_second_step_detail.R	Extended tables/figures → Second Step/analysis/

Models: **composer-2** (Second Step), **opus-4.6** (Third Step), **grok-4.6** (in progress). Agents are sandboxed with `settingSources: []` so user/project rules outside the workspace are not loaded.

5 4. Key results — Composer-2 run (Second Step, $N = 250$)

5.1 Arm-level ATT summary

Table 6: Composer-2 arm-level ATT summary (Second Step).

Arm	N	Mean ATT	SD	Median	% neg.	% teen	% TWFE	Mean span
none	50	-0.0050	0.0165	-0.0091	78.0	100.0	0.0	26.5
null_short	50	-0.0064	0.0143	-0.0080	80.0	100.0	2.0	25.7
null_cites	50	-0.0002	0.0209	-0.0059	70.0	100.0	0.0	23.3
negative_short	50	-0.0045	0.0137	-0.0082	74.0	98.0	4.0	25.5
negative_cites	50	-0.0053	0.0187	-0.0098	76.0	100.0	0.0	27.4

5.2 Inference

- **Fisher omnibus** (between-arm sum of squares, 10,000 permutations): observed = 0.0011, $p = 0.4101$.

- **Kruskal–Wallis** across five arms: $\chi^2 = 4.8195$, $p = 0.3063$.
- Planned contrasts (null – negative, short and cites): neither rejects at conventional levels.

Interpretation: condition labels are statistically exchangeable with respect to reported ATT in the composer-2 run.

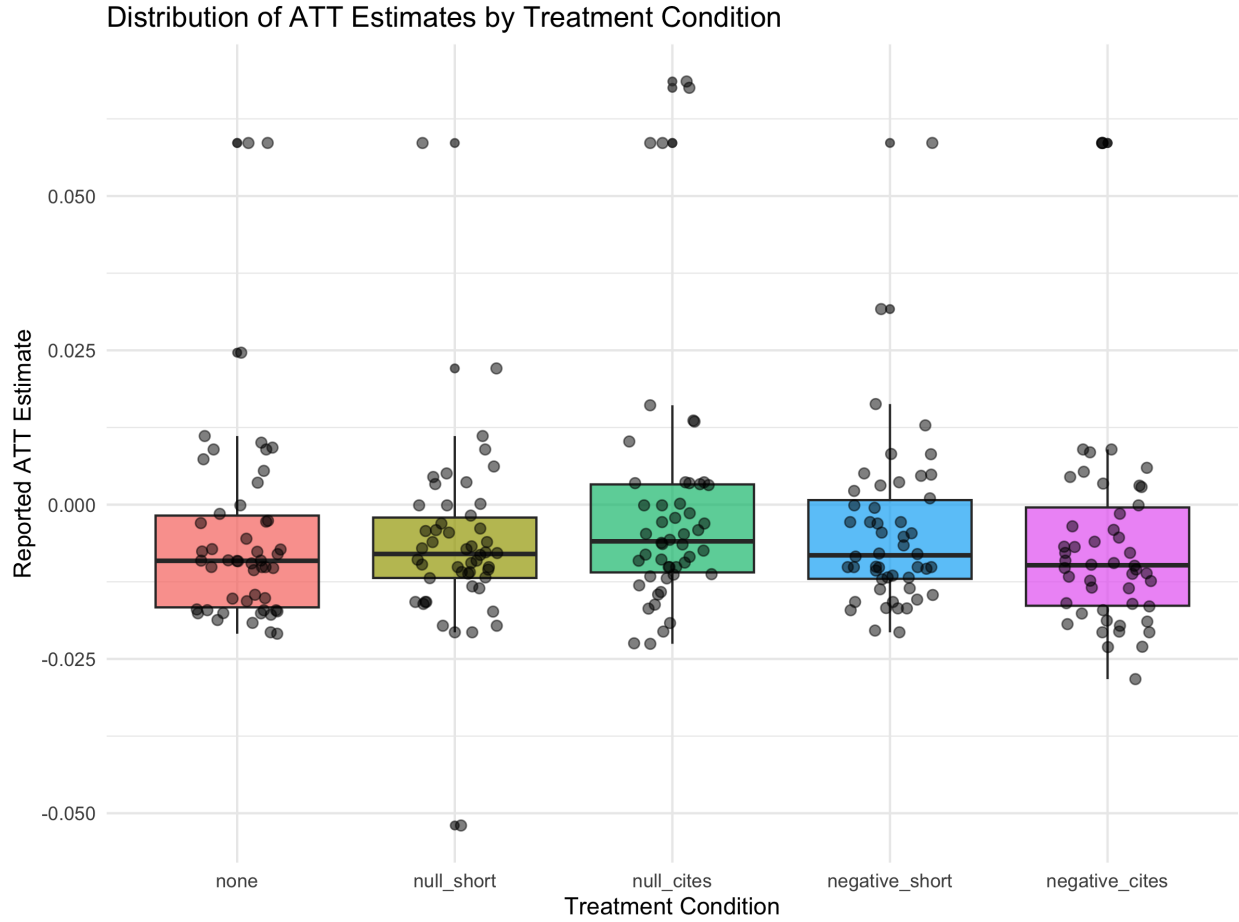


Figure 1: Distribution of reported ATT estimates by treatment arm — Composer-2 (N = 250).

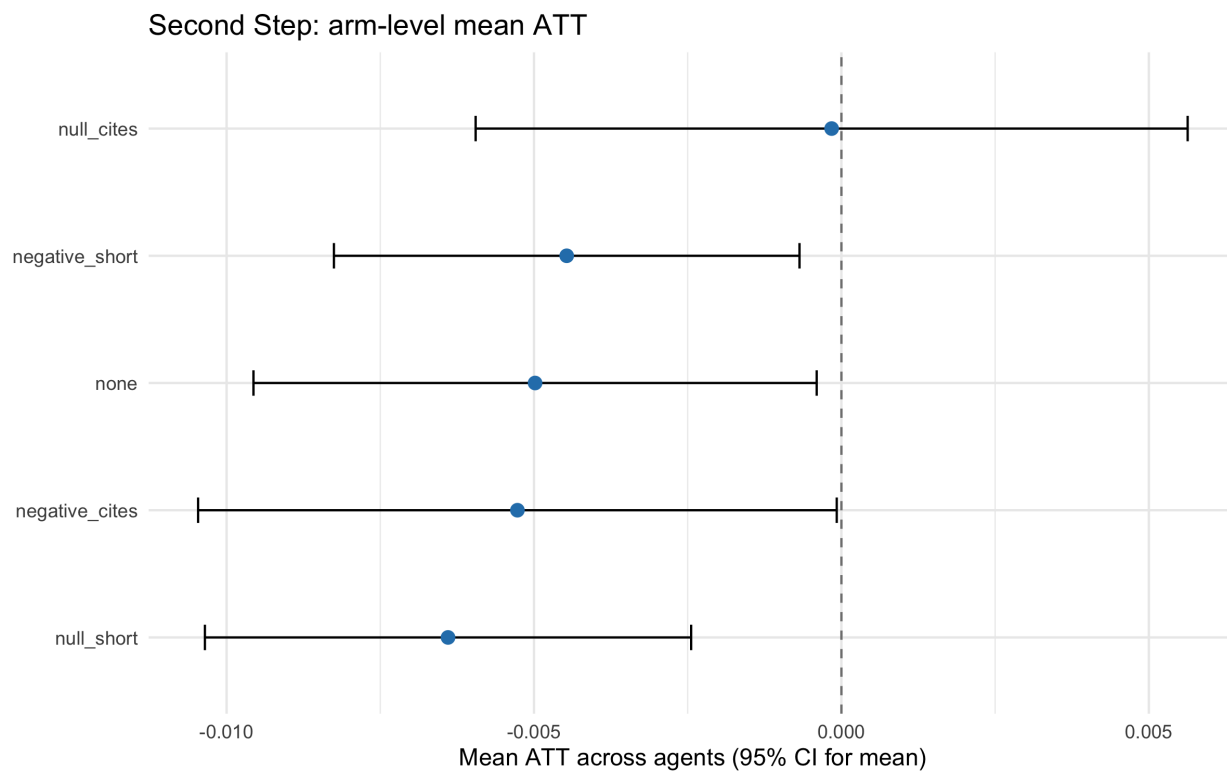


Figure 2: Arm-level mean ATT with approximate 95% intervals — Composer-2.

5.3 Specification patterns (Composer-2)

- **Outcome:** nearly universal teen EMP/POP (`emp_pop_ratio_teen`).
- **Estimator:** Callaway–Sant’Anna dominates; TWFE rare and not concentrated in negative arms.
- **Treatment:** almost entirely binary staggered MW changes.
- **Covariates:** nontrivial share use none (~42–66% by arm); not clearly sorted by prime.
- **Event studies:** produced by 98% of agents per arm.

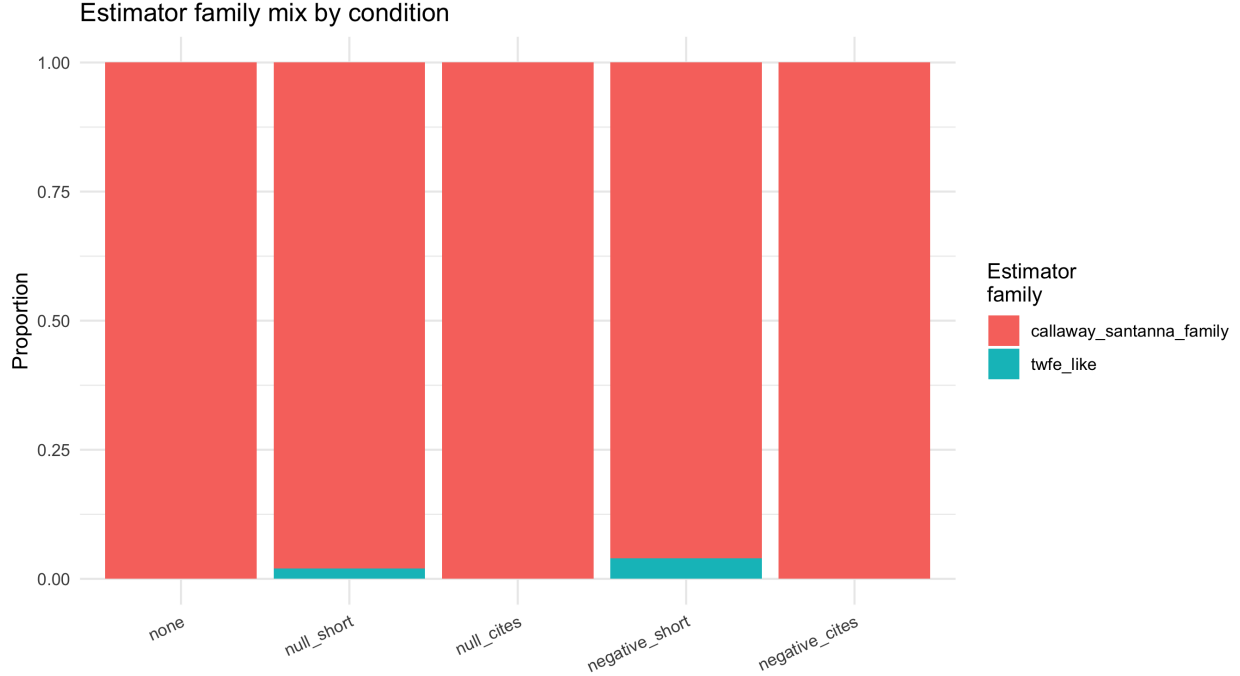


Figure 3: Estimator family composition by arm — Composer-2.

6 5. Key results — Opus 4.6 run (Third Step, $N = 249$)

The Opus 4.6 run used the **identical protocol, data, and instructions** as the Composer-2 Second Step, differing only in the underlying AI model (claude-opus-4-6 via Cursor SDK). One agent failed, yielding 249 usable results (49 in `negative_cites`, 50 in all other arms).

6.1 Arm-level ATT summary

Table 7: Opus 4.6 arm-level ATT summary (Third Step).

Arm	N	Mean ATT	SD	Median	% neg.	% teen	% TWFE	% continuous	Mean span
none	50	0.0035	0.0102	0.0037	34.0	100.0	0.0	0.0	13.9
null_short	50	0.0003	0.0123	0.0010	48.0	100.0	0.0	0.0	16.3
null_cites	50	0.0055	0.0085	0.0085	20.0	98.0	0.0	0.0	14.0
negative_short	50	-0.0027	0.0137	-0.0085	68.0	94.0	2.0	6.0	14.4
negative_cites	49	0.0047	0.0106	0.0076	34.7	100.0	0.0	2.0	13.2

6.2 Inference

- ****Fisher omnibus**** (between-arm SS, 10,000 permutations): observed = 0.0023, *p* = 0.0010
- ****Kruskal-Wallis**** (all five arms): chi-squared = 13.9947, *p* = 0.0073

The between-arm difference is statistically significant. The `negative_short` arm is the clear outlier: mean ATT = -0.0027, median = -0.0085 — the most negative of all five arms.

Pairwise comparisons (Wilcoxon rank-sum, unadjusted):

Table 8: Pairwise Wilcoxon rank-sum tests (Opus 4.6 run, unadjusted).

Comparison	p-value
negative_short vs null_cites	0.0024
negative_cites vs negative_short	0.0100
null_short vs null_cites	0.0192
null_cites vs null_short	0.0210
null_cites vs negative_short	0.0711
null_short vs null_short	0.1656
null_short vs none	0.1766
null_short vs negative_short	0.4382
null_cites vs null_cites	0.4654
negative_short vs negative_short	0.7386

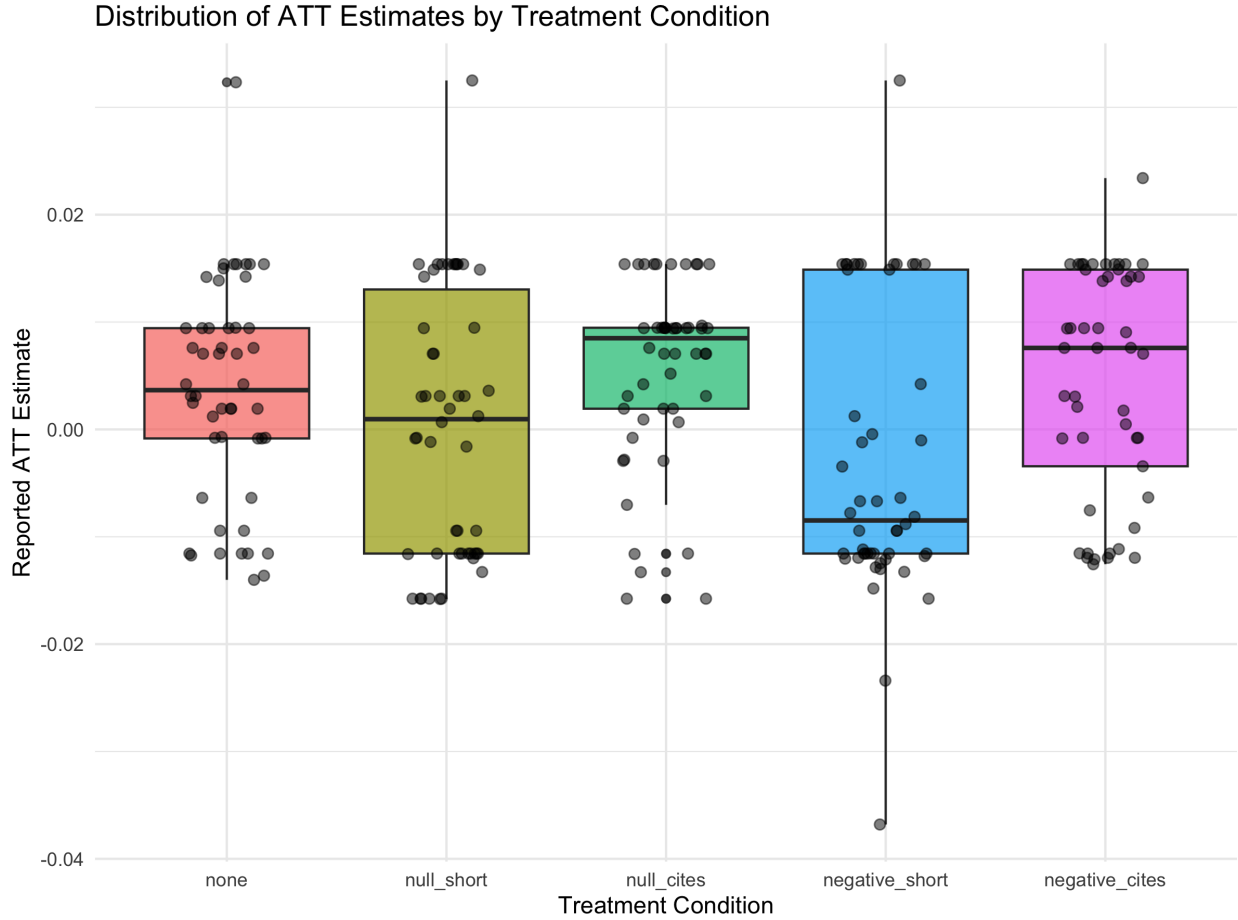


Figure 4: Distribution of reported ATT estimates by treatment arm — Opus 4.6 (N = 249).

6.3 Specification patterns (Opus 4.6)

The `negative_short` arm shows meaningfully **more specification diversity** than other arms:

- **3 agents** used **continuous treatment** definitions (vs 0 in other arms).

- **2 agents** used **CGBS** / **continuous DiD** estimators.
- **1 agent** used **TWFE**.
- **11 agents** used **never-treated** as the control group (vs predominantly not-yet-treated in other arms).
- The remaining agents used Callaway–Sant’Anna, consistent with the overall pattern.

This specification variation in the **negative_short** arm is reminiscent of Cunningham’s **Wave 2** finding: negatively primed agents exercising more estimator discretion. A Fisher exact test of treatment definition (binary vs continuous) $\times$ arm yields $p = 0.051$ (borderline).

7 6. Cross-model comparison

Table 9: Cross-model comparison: arm-level ATT and specification choices.

Model	Arm	N	Mean ATT	SD	% TWFE	% continuous Tx
Composer-2	none	50	-0.0050	0.0165	0.0	0.0
Composer-2	null_short	50	-0.0064	0.0143	2.0	0.0
Composer-2	null_cites	50	-0.0002	0.0209	0.0	0.0
Composer-2	negative_short	50	-0.0045	0.0137	4.0	0.0
Composer-2	negative_cites	50	-0.0053	0.0187	0.0	0.0
Opus 4.6	none	50	0.0035	0.0102	0.0	0.0
Opus 4.6	null_short	50	0.0003	0.0123	0.0	0.0
Opus 4.6	null_cites	50	0.0055	0.0085	0.0	0.0
Opus 4.6	negative_short	50	-0.0027	0.0137	2.0	6.0
Opus 4.6	negative_cites	49	0.0047	0.0106	0.0	2.0

Table 10: Omnibus tests across model runs. Grok 4.6 is in progress.

Model	N	Fisher p	KW p	Significant?
Composer-2	250	0.4101	0.3063	No
Opus 4.6	249	0.0010	0.0073	Yes
Grok 4.6	NA	—	—	Pending

Key contrast: The same experimental protocol yields **no detectable between-arm differences** with **composer-2** but **significant differences** with **opus-4.6**. The divergence is concentrated in the **negative_short** arm, which under Opus 4.6 produces more negative ATTs and more varied specifications — paralleling Cunningham’s finding that negatively primed agents under certain conditions move toward TWFE/continuous treatment and more negative estimates.

8 7. Comparison to Cunningham (2026)

Dimension	Cunningham (Substack)	Composer-2 run	Opus 4.6 run
Platform	Claude Code CLI	Cursor SDK (composer-2)	Cursor SDK (opus-4.6)
Arms	3	5	5

Dimension	Cunningham (Substack)	Composer-2 run	Opus 4.6 run
Wave 1 pattern	Similar ATTs under CS-only	Replicates (~99% CS; Fisher $p = 0.41$)	Partial (mostly CS, but arm differences emerge)
Wave 2 pattern	Negative arm $\rightarrow$ more TWFE, continuous Tx, more negative ATTs	Does not replicate (4% TWFE; 0% continuous)	Partially replicates (negative_short arm shifts negative with specification diversity)
Data	Same essential panel concept	Pre-aggregated CSV	Pre-aggregated CSV

The Opus 4.6 run **bridges the gap** between the Composer-2 results and Cunningham’s Wave 2 findings. While the specification-channel effect is concentrated in the **negative_short** arm (rather than all negative-primed agents), the pattern is qualitatively similar: negative primes induce some agents to explore alternative estimators and report more negative ATTs.

Most plausible explanation: **model matters**. **Composer-2** agents treat TWFE caveats as binding and default uniformly to CS. **Opus-4.6** agents exercise more discretion when primed with negative literature, especially in the concise (**short**) format. A fuller technical discussion is in `reports/cursor_replication_report.qmd`.

9 8. Repository map

```

First Step/                                # Frozen first-wave materials
Second Step/
  results/                                # 250-agent Composer-2 outputs + core plots
  analysis/                              # Extended summaries and figures
Third Step/
  results_opus46/                        # 249-agent Opus 4.6 outputs + core plots
  results_grok46/                        # Grok 4.6 run (in progress)
data/agent_panel_essential.csv
conditions.tsv
INSTRUCTIONS_SHARED.md / DID_METHODODOLOGY.md / PRIME_*.md
scripts/                                # launch, collect, analyze
PROJECT_LOG.md                          # Implementation history and troubleshooting
reports/                                # This summary + full replication report

```

Reproduce analysis:

```

Rscript scripts/analyze_results.R "Second Step/results"
Rscript scripts/analyze_second_step_detail.R
Rscript scripts/analyze_results.R "Third Step/results_opus46"

```

10 9. Limitations and next steps

Limitations. Two models completed so far (`composer-2` and `opus-4.6`); Grok 4.6 run in progress. No rhetoric/confidence scoring pipeline; one permissive-estimator wave on Cursor per model (no within-platform CS-only vs TWFE-allowed contrast); prime texts differ from Cunningham’s exact wording.

Testable next steps.

1. **Complete Grok 4.6 run** and incorporate results for a three-model comparison.
 2. Match Cunningham’s three-arm primes and Wave 2 methodology text verbatim.
 3. Inspect Opus 4.6 agent logs for the **negative_short** arm to understand the specification-search mechanism.
 4. Re-run with additional models via Cursor SDK to map the model → prime-sensitivity frontier.
 5. Optionally relax sandbox (**settingSources**) to test ambient-memory hypotheses.
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11 References

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- Borjas, G. J., & Breznau, N. (2026). Many-analyst study of immigration. *Science Advances*.
- Baker, A., Callaway, B., Cunningham, S., Goodman-Bacon, A., & Sant’Anna, P. H. C. (2025). Difference-in-Differences Designs: A Practitioner’s Guide.
- Sant’Anna, P. H. C. (2025). *My Claude Code Setup*. Emory University. <https://psantanna.com/claude-code-my-workflow/>

12 Acknowledgments

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